

Cloud Computing Assignment – Task 3

Cloud-Based Bus Pass System

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Module: Cloud Computing

Application: Cloud-Based Bus Pass System

Programming Language: Python

Framework: Flask

1. Introduction

The purpose of this project was to develop a cloud-based web application that allows users to register, log in, browse available bus routes, book bus tickets, and manage their bookings.

The application was developed using the Flask framework with SQLite as the database. Passwords are securely stored using Bcrypt hashing, and the system also provides an administrator panel for managing bus routes.

2. Objectives

The objectives of the project were:

- Develop a cloud-based web application using Flask.
- Create a secure user registration and login system.
- Store application data in a relational database.
- Allow users to browse available bus routes.
- Enable users to book bus tickets.
- Generate unique ticket numbers.
- Allow users to cancel bookings.
- Provide an administrator interface for managing bus routes.

3. Technologies Used

The following technologies were used during the development of the application:

- Python 3
- Flask
- Flask-SQLAlchemy
- Flask-Bcrypt
- SQLite
- HTML5
- CSS3
- Jinja2 Templates
- Visual Studio Code
- Git & GitHub

4. System Design

The application follows the Flask MVC architecture.

The system consists of the following components:

Frontend

- HTML
- CSS
- Jinja Templates

Backend

- Flask
- Python

Database

SQLite database containing:

- Users
- Routes
- Bookings

The application communicates with the database using SQLAlchemy ORM.

5. Features Implemented

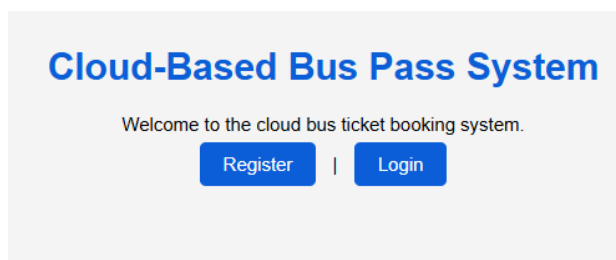
The application includes the following functionality.

User Registration

Users can create a new account using their full name, email address, and password.

Passwords are encrypted using Flask-Bcrypt before being stored in the database.

Figure 1 – Registration page

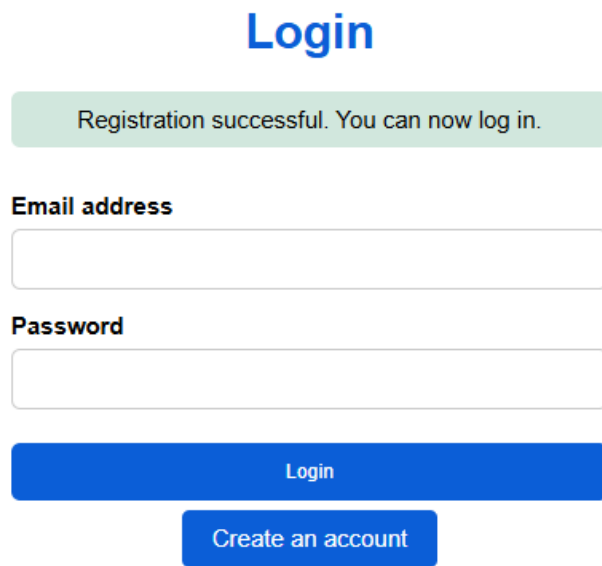


Secure Login

Registered users can securely log into the system.

Incorrect login details generate an error message.

Figure 2 – Login page



The login page features a large blue 'Login' title at the top. Below it is a light green success message box stating 'Registration successful. You can now log in.' The form includes two input fields: 'Email address' and 'Password'. At the bottom, there are two blue buttons: 'Login' and 'Create an account'.

Login

Registration successful. You can now log in.

Email address

Password

Login

Create an account

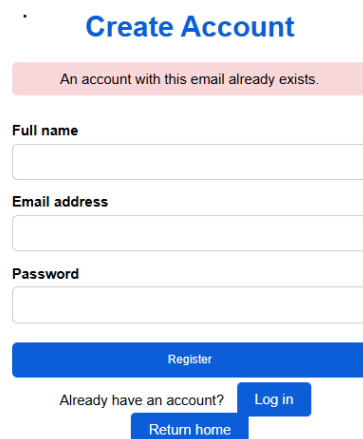
Dashboard

After logging in, users are redirected to the dashboard.

The dashboard provides access to:

- Available Routes
- My Bookings
- Admin Panel
- Logout

Figure 3 – User Dashboard



The 'Create Account' page has a blue title. A pink error message box states 'An account with this email already exists.' The form contains three input fields: 'Full name', 'Email address', and 'Password'. At the bottom, there is a blue 'Register' button, and below it, the text 'Already have an account?' followed by a blue 'Log in' button and a blue 'Return home' button.

Create Account

An account with this email already exists.

Full name

Email address

Password

Register

Already have an account? Log in

Return home

Available Routes

Users can browse all available bus routes including:

- Origin
- Destination
- Departure Time
- Ticket Price
- Available Seats

Figure 4 – Available Routes

The screenshot displays a web interface titled "Available Bus Routes". At the top, a green notification bar states "Login successful.". Below this, four bus routes are listed, each with its origin and destination, departure time, price, available seats, and a "Book ticket" button. The routes are: London to Manchester, London to Birmingham, Leeds to Liverpool, and Glasgow to Edinburgh. At the bottom of the page, there is a "Back to dashboard" button.

Route	Departure	Price	Available seats	Action
London → Manchester	08:00	£24.99	40	Book ticket
London → Birmingham	10:30	£18.50	35	Book ticket
Leeds → Liverpool	12:15	£16.75	30	Book ticket
Glasgow → Edinburgh	15:00	£12.99	50	Book ticket

Back to dashboard

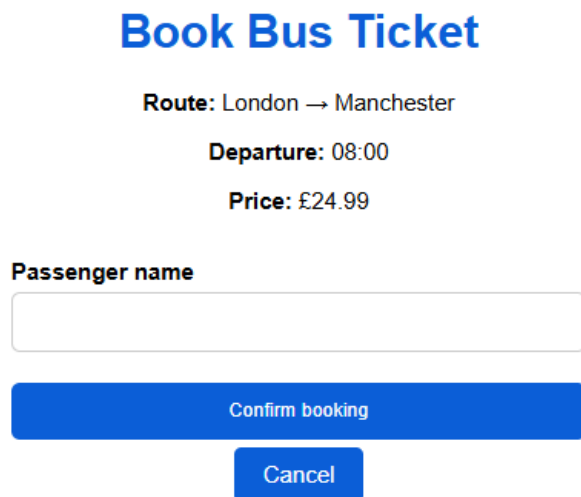
Ticket Booking

Users can book a ticket by entering the passenger name.

After booking:

- available seats decrease,
- booking information is stored,
- a unique ticket number is generated.

Figure 5 – Booking Form



Book Bus Ticket

Route: London → Manchester

Departure: 08:00

Price: £24.99

Passenger name

Confirm booking

Cancel

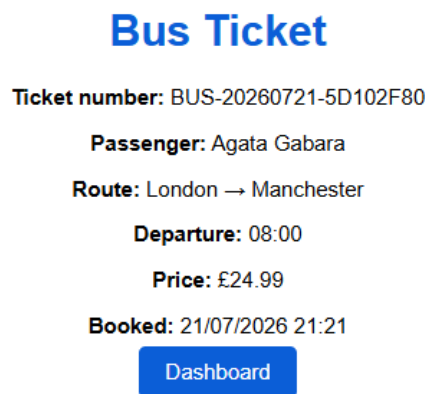
Ticket Generation

After a successful booking, the application generates a ticket containing:

- Ticket Number
- Passenger Name
- Route
- Departure Time
- Price

- Booking Date

Figure 6 – Generated Ticket



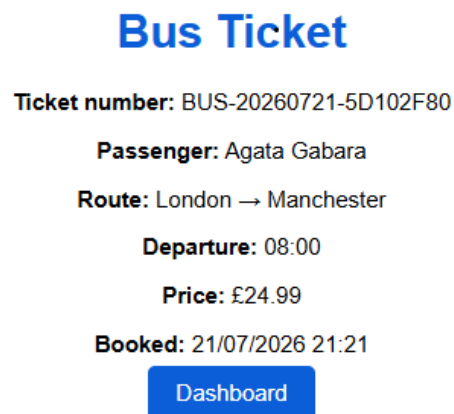
Booking Management

Users can:

- view booking history,
- cancel bookings.

Cancelling a booking automatically increases the number of available seats.

Figure 7 – My Bookings



Administrator Panel

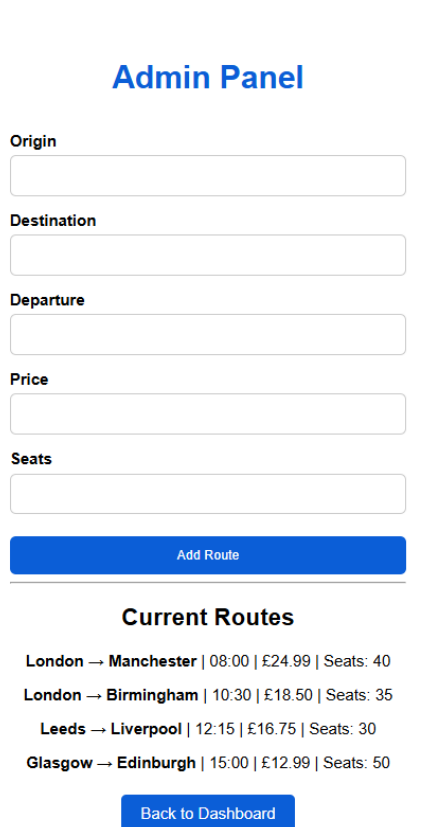
The administrator panel allows adding new bus routes.

The administrator enters:

- Origin
- Destination
- Departure Time
- Ticket Price
- Number of Seats

The newly created route becomes immediately available for booking.

Figure 8 – Administrator Panel



The screenshot shows a web interface titled "Admin Panel" in blue. It contains five input fields for "Origin", "Destination", "Departure", "Price", and "Seats", each with a label above it. Below these fields is a blue button labeled "Add Route". Underneath the button is a section titled "Current Routes" which lists four routes: "London → Manchester | 08:00 | £24.99 | Seats: 40", "London → Birmingham | 10:30 | £18.50 | Seats: 35", "Leeds → Liverpool | 12:15 | £16.75 | Seats: 30", and "Glasgow → Edinburgh | 15:00 | £12.99 | Seats: 50". At the bottom of this section is a blue button labeled "Back to Dashboard".

Admin Panel

Origin

Destination

Departure

Price

Seats

Add Route

Current Routes

London → Manchester | 08:00 | £24.99 | Seats: 40
London → Birmingham | 10:30 | £18.50 | Seats: 35
Leeds → Liverpool | 12:15 | £16.75 | Seats: 30
Glasgow → Edinburgh | 15:00 | £12.99 | Seats: 50

Back to Dashboard

Route Added Successfully

The application displays a confirmation message after adding a new route.

Figure 9 – Route Successfully Added

Admin Panel

Route added successfully.

Origin

Destination

Departure

Price

Seats

Add Route

Current Routes

London → Manchester | 08:00 | £24.99 | Seats: 40

London → Birmingham | 10:30 | £18.50 | Seats: 35

Leeds → Liverpool | 12:15 | £16.75 | Seats: 30

Glasgow → Edinburgh | 15:00 | £12.99 | Seats: 50

Oxford → Cambridge | 09:30 | £2250.00 | Seats: 25

Back to Dashboard

6. Database Structure

The application uses SQLite.

The database contains three main tables.

User

- id
- full_name
- email
- password_hash

Route

- id
- origin
- destination
- departure_time
- price
- available_seats

Booking

- id
- ticket_number
- passenger_name
- booking_date
- total_price
- user_id
- route_id

Relationships:

- One User can have many Bookings.
- One Route can have many Bookings.

7. Testing

The application was tested using multiple scenarios.

Registration

- ✓ New account created successfully.
- ✓ Duplicate email prevented.

Login

- ✓ Correct credentials allowed access.
- ✓ Incorrect password rejected.

Booking

- ✓ Ticket generated successfully.
- ✓ Seats reduced automatically.

Booking Cancellation

- ✓ Booking removed.
- ✓ Available seats restored.

Administrator Panel

- ✓ New routes added successfully.
- ✓ New routes immediately visible in the Available Routes page.

No critical errors were identified during testing.

8. Challenges

Several challenges were encountered during development.

- Resolving circular imports between Flask application files.
- Managing user sessions.
- Secure password hashing using Flask-Bcrypt.
- Creating relationships between database tables.

- Automatically updating available seats after booking and cancellation.
- Implementing the administrator panel.

These issues were successfully resolved.

9. Conclusion

The Cloud-Based Bus Pass System successfully meets the project objectives.

The application provides secure user authentication, bus ticket booking, booking management, and administrator functionality.

The project demonstrates practical use of Flask, SQLAlchemy, SQLite, and modern web development techniques.

The application is reliable, user-friendly, and can be extended with additional functionality such as payment processing, email ticket confirmation, and administrator authentication.

10. Screenshots

Insert the following screenshots in order:

- Figure 1 – Registration Page
- Figure 2 – Login Page
- Figure 3 – Dashboard
- Figure 4 – Available Routes
- Figure 5 – Booking Form
- Figure 6 – Generated Ticket
- Figure 7 – My Bookings
- Figure 8 – Administrator Panel
- Figure 9 – Route Added Successfully